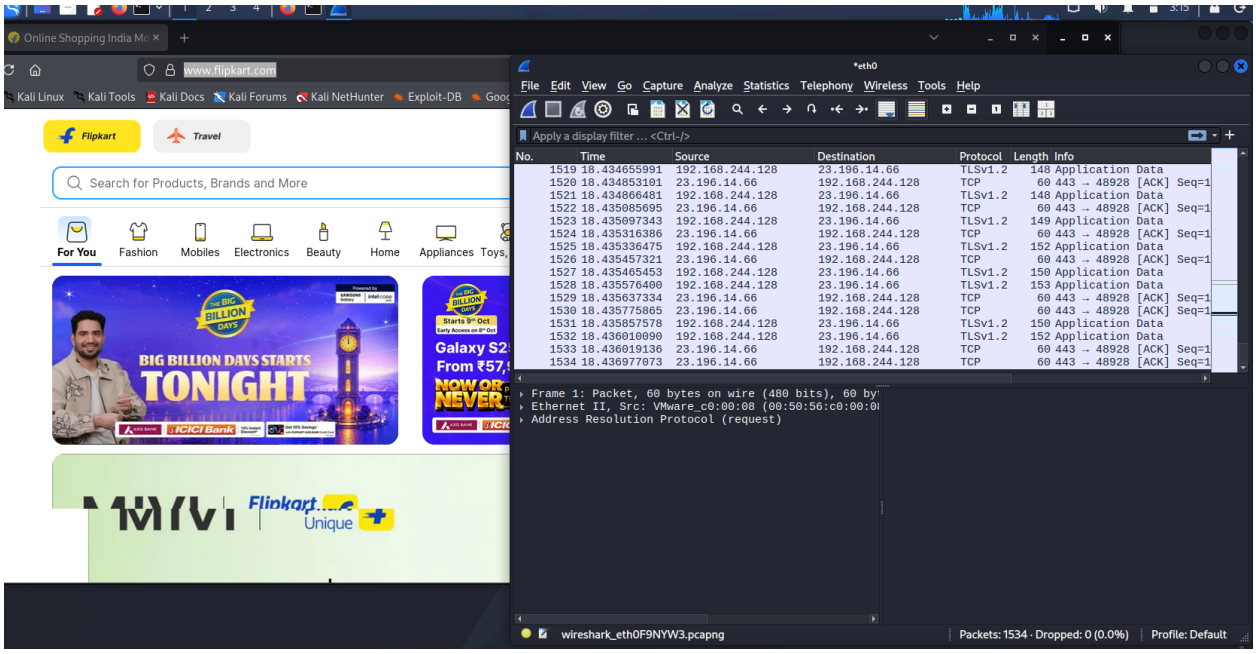
1.Open Wireshark, capture network traffic while you refresh the Flipkart homepage, and identify one packet each examining belonging to the Application, Transport, Internet, and Network Access layers by their headers.



Protocol names in packet list:

- **TLSv1.2 (Application Data packets)**
- **TCP (the [ACK] packets, 443 → 48928)**

1. **Application Layer:**

   TLSv1.2 (Transport Layer Security)

2. **Transport Layer:**

   TCP (Transmission Control Protocol)

3. **Internet Layer:**

   IPv4 (Inferred from the Source/Destination IP addresses like 192.168.244.128)

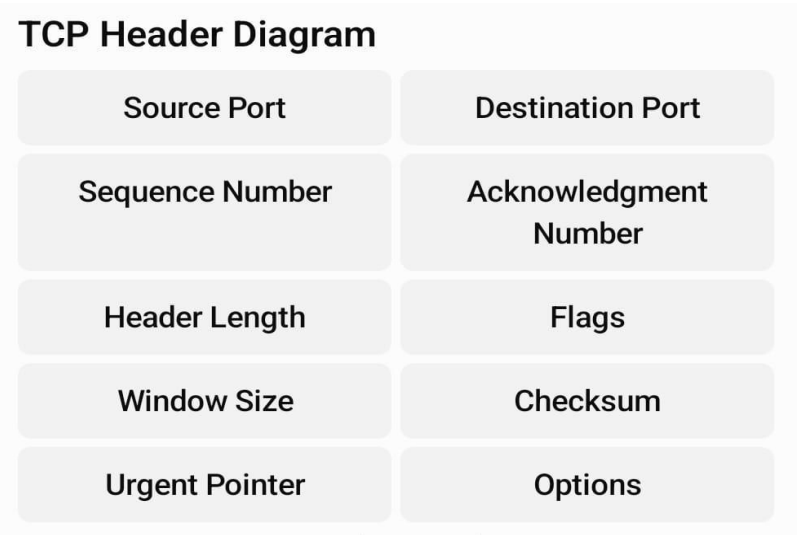
4. **Network Access Layer:**

   Ethernet II (Visible under the packet details pane at the bottom)

2.Create a table comparing the OSI and TCP/IP models, listing all layer side-by-side, and for each layer, give a real-world example from any app you use (e.g., WhatsApp message, Zomato order).

OSI Model	TCP/IP Model	Real-world example
7. Application	**Application**	WhatsApp: typing and sending a message
6. Presentation	**Application**	WhatsApp: message encryption/formatting
5. Session	**Application**	WhatsApp: maintaining the communication session
4. Transport	**Transport**	Message is broken into data pieces and delivered
3. Network	**Internet**	IP address helps the message reach the receiver
2. Data Link	**Network Access**	Wi-Fi/Ethernet sends frames between devices
1. Physical	**Network Access**	Wi-Fi radio signals or Ethernet electrical signals

3.Given a screenshot of a TCP segment header (provided by your trainer), label the fields that belong to the Transport layer and explain what each field does in 1-2 lines.

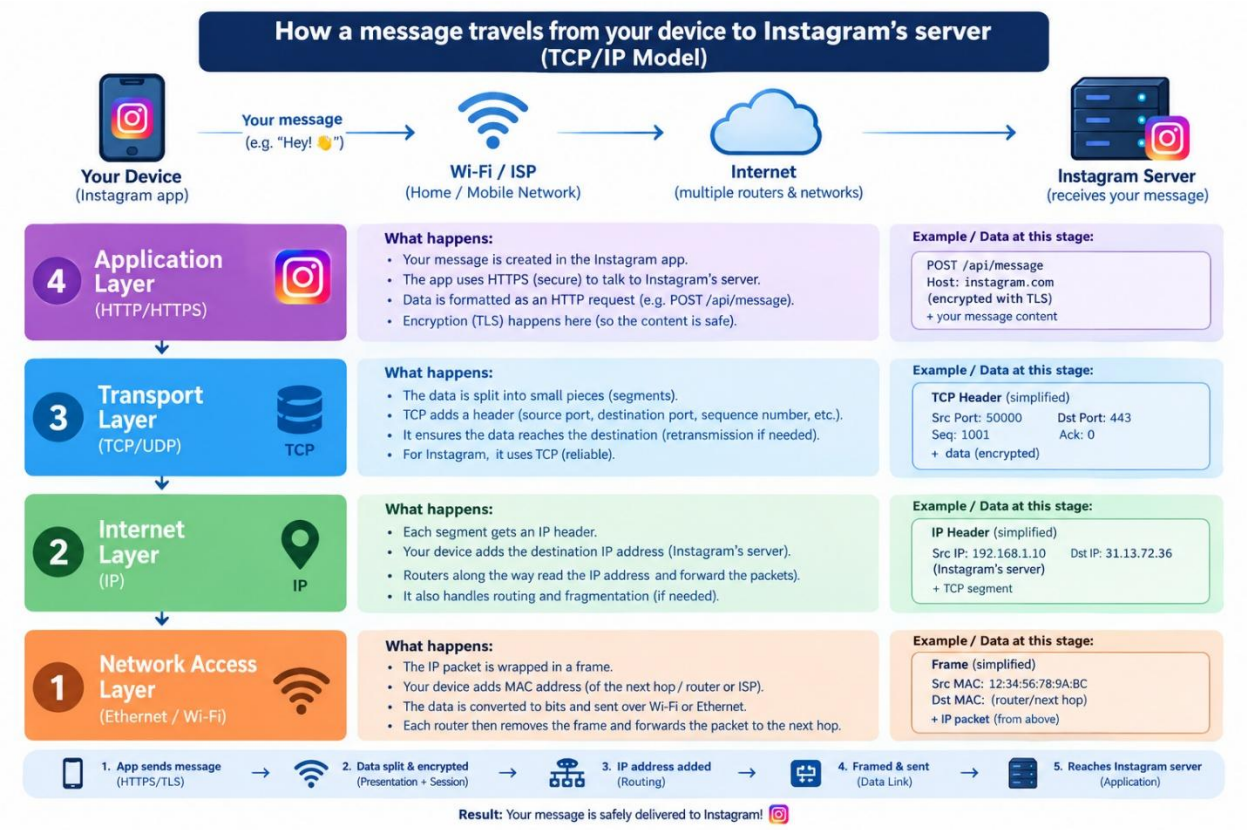


TCP Segment Header Fields:

TCP Field	Function
1. Source Port	Identifies the sender's application.
2. Destination Port	Identifies the receiver's application.
3. Sequence Number	Keeps data in the correct order.

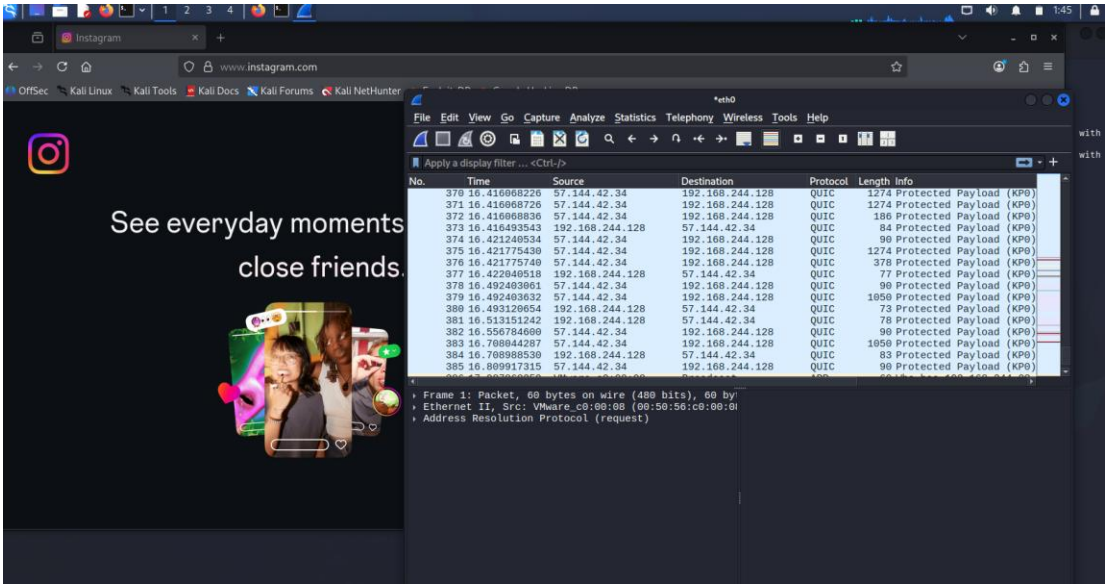
TCP Field	Function
4. Acknowledgment Number	Confirms received data.
5. Header Length	Shows the size of the TCP header.
6. Flags	Controls connection setup and closing.
7. Window Size	Controls how much data can be received.
8. Checksum	Checks for errors in data.
9. Urgent Pointer	Identifies urgent data.
10. Options	Contains extra TCP settings.

4.Draw a diagram showing how a message sent from your device to Instagram's server passes through each TCP/IP layer, and annotate what happens at each step using simple app-related terms (e.g., 'message split into packets', 'IP address added', etc.).



5.Pick any one packet you captured in Wireshark and write which TCP/IP layer it belongs to, how you identified it, and what real-world action in an app (like sending a WhatsApp message or booking a ticket on BookMyShow) it represents.

Capture Instagram web-app traffic



1. **TCP/IP Layer: Transport Layer (QUIC uses UDP)**

2. **How I identified it: I captured Instagram traffic in Wireshark and saw the protocol name QUIC in the Protocol column. QUIC works over UDP and provides fast, secure communication.**

3. **Real-world action: It represents Instagram app traffic, such as loading posts, reels, or other content from Instagram's**

servers. My capture shows QUIC traffic, but it does not prove that a specific message was sent.